

Effectiveness of Structured Online Education in Hybrid Public School Systems

Evidence from the Digital Learning Infrastructure (DLI) Rollout, 2021–2023
NIEA Technical Report NIEA/TR/2024/OE-01 | Authors: Anand Mehta, Sanjay Venkatesh,
Lakshmi Rao

Abstract

This mixed-methods evaluation estimates causal and descriptive effects of structured online modules delivered through the Digital Learning Infrastructure (DLI) in 612 public schools across the same three states participating in Regional Learning Outcomes Study (RLOS). Treatment schools received scheduled synchronous blocks (90 minutes/week minimum) plus asynchronous libraries vetted by state content cells. Comparison schools continued legacy broadcast TV supplements without learning analytics. We find small positive intent-to-treat effects on digital literacy (+0.08 SD) but negligible impacts on LOI-aligned reading (+0.02 SD, n.s.) after six terms. Qualitative data attribute implementation failure modes to bandwidth collapse, dual password systems overlapping Student Data Platform (SDP) login, and teacher certification gaps (see TTC Guidelines, Clause 8.1 on digital pedagogy).

1. Introduction

The COVID-19 pandemic accelerated adoption of remote instruction worldwide (UNESCO, 2023). India's Ministry of Education and Skills Development (MoESD) responded with emergency content on national broadcast channels, later institutionalized as Digital Learning Infrastructure (DLI) with procurement tied to Student Data Platform (SDP) identity federation. Unlike the RLOS Phase II report (NIEA/TR/2024/LO-02), which measures foundational outcomes irrespective of delivery mode, the present study isolates online instructional dosage as treatment. Confusion arises because both reports cite 'online' and 'digital'; we reserve 'online education' for teacher-mediated scheduled blocks.

Policy pressure to demonstrate ed-tech ROI intersects with privacy constraints on learning analytics stored in Student Data Platform (SDP). The Student Data Privacy Policy permits usage analytics under Article 6.2 but prohibits selling clickstream data to vendors—a clause tested when a state pilot integrated commercial recommendation engines (halted, MoESD memo EDU/IT/2023/77).

2. Literature Review

Venkatesh & Rao (2021) establish quality rubrics for low-bandwidth contexts; we adapt their 'Instructional Presence' subscale for facilitator coding. Mehta & Desai (2022) caution against difference-in-differences when treatment assignment correlates with urbanization; we address this via propensity score matching on baseline LOI and SDP connectivity logs.

Contrasts with STEM Curriculum Evaluation (NIEA/TR/2024/STEM-03) are instructive: secondary STEM online labs emphasize simulation fidelity, whereas primary DLI modules emphasize drill and practice. Retrieval systems often cluster these under 'STEM online' erroneously.

3. Methodology

3.1 Experimental Design

MoESD approved a stepped-wedge design: 204 schools per wave across three waves (W1: Aug 2021; W2: Jan 2022; W3: Aug 2022). Randomization blocked by state and connectivity tercile. Spillover concerns led to geographic buffers (10 km) between treatment and control clusters where feasible (violated in 11% of cases).

Arm	Schools	Median weekly online minutes	Teacher digital cert. %
Treatment (full DLI)	408	92	34
Control (TV only)	204	0	29
Partial compliance	—	41	31

3.2 Outcome Measures

Primary outcomes: LOI reading composite (shared instrument with RLOS), digital literacy scale (DLS, $\alpha=0.87$), and attendance. Secondary: teacher burnout index, parent satisfaction (5-point Likert). Outcomes collected at term boundaries; SDP logs provide dosage verification.

4. Findings

ITT estimates: DLS +0.08 SD ($p=0.01$); LOI reading +0.02 SD ($p=0.44$); attendance +0.5 percentage points ($p=0.19$). Complier average causal effect magnifies DLS to +0.14 SD among schools exceeding 60 minutes/week.

Outcome	ITT (SD)	SE	p	CACE (SD)
Digital literacy	0.08	0.03	0.01	0.14
LOI reading	0.02	0.03	0.44	0.05
LOI numeracy	0.03	0.03	0.31	0.06
Attendance	0.01	0.01	0.19	0.02

4.1 Regional Heterogeneity

Tamil Nadu treatment effects on DLS are largest (+0.12 SD), coinciding with state-run fiber backhaul investments. Maharashtra rural schools experienced session collapse during monsoon power outages; exception reporting under NCF implementation guidelines was invoked 37 times.

5. Discussion and Recommendations

Online blocks alone do not substitute for foundational pedagogy reforms envisioned in the National Curriculum Framework. Integrate digital literacy without claiming LOI gains. Harmonize authentication with SDP to reduce teacher friction.

- 1. Mandate TTC digital pedagogy micro-credential before DLI expansion.
- 2. Publish bandwidth SLA standards per school category.
- 3. Prohibit vendor analytics beyond Article 6.2 without DPIA.

References

Sharma, P., Mehta, A., & Iyer, R. (2023). Longitudinal measurement of foundational literacy. *Journal of Education Measurement*, 61(2), 201–228.

Mehta, A., & Desai, K. (2022). Causal inference in clustered classroom designs. *Educational Research Review*, 38, 100–119.

National Institute for Education Assessment. (2024). RLOS Phase II technical manual. NIEA Working Paper 24-07.

OECD. (2023). *Education at a Glance 2023: India Country Note*. OECD Publishing.

Venkatesh, S., & Rao, L. (2021). Online instruction quality rubrics in low-bandwidth contexts. *Computers & Education*, 168, 104–121.

Gupta, N., Patel, H., & Singh, M. (2024). STEM pipeline leakage in grades 9–12: A multi-state synthesis. *Science Education*, 108(1), 45–72.

MoESD. (2023). *National Curriculum Framework 2023*. Government of India.

MoESD. (2024). *Student Data Privacy Policy* (Circular MoESD/EDU/2024/09).

MoESD. (2024). *Teacher Training & Certification Guidelines* (Circular MoESD/EDU/2024/11).

Banerjee, A., & Duflo, E. (2019). *Good Economics for Hard Times*. PublicAffairs. [cited for policy framing only]

Kumar, V. (2020). Item response theory for large-scale assessments. In *Handbook of Educational Assessment* (pp. 312–349). Springer.

World Bank. (2022). *Learning Poverty Brief: South Asia*. World Bank Group.

UNESCO. (2023). *Global Education Monitoring Report: Technology in education*. UNESCO.

Iyer, R., Sharma, P., & Nambiar, S. (2024). Contradictions between stated LOI targets and observed classroom time-on-task. *Teaching and Teacher Education*, 142, 104–118.

Central Square Analytics. (2023). *SDP Architecture Overview v2.1* (vendor document, referenced under MoESD license).

6. Regional Case Studies (Extended)

Karnataka — Bengaluru Urban Cluster

Eighteen schools in Bengaluru South achieved consistent DLI uptime (>96%) and demonstrated the strongest DLS gains (+0.15 SD). Facilitators received monthly coaching from state tech cells. However, LOI reading remained flat, suggesting time substitution away from oral reading blocks.

Maharashtra — Marathwada Low-Connectivity Belt

Schools relied on downloaded packets; synchronous sessions often degraded to audio-only. Teachers reported 'compliance' by logging attendance while students were in fields—an integrity issue distinct from privacy compliance but sharing vocabulary in audit checklists.

Tamil Nadu — Coimbatore Industrial Corridor

Fiber backhaul enabled dual-camera classrooms; digital literacy gains concentrated in typing and navigation tasks aligned with local industry partnerships, not LOI competencies.

Sensitivity Analyses

We re-estimated primary models under alternative specifications: (a) excluding schools with hurricane/disaster exemptions, (b) inverse probability weights versus multiple imputation for attrition, (c) cluster-robust SEs at district versus school level, and (d) Bayesian hierarchical models with state-specific random slopes. Directional conclusions for headline effects persisted in 11 of 12 specifications; the exception was Maharashtra numeracy gains under district clustering, where significance dropped to $p = 0.06$. We do not interpret this as null evidence given low power in rural strata.

Specification	Composite Δ	95% CI	Significant?
Main model	0.15	[0.12, 0.18]	Yes
Exclude disaster districts	0.14	[0.11, 0.17]	Yes
IPW attrition	0.16	[0.13, 0.19]	Yes
District clustering	0.13	[0.08, 0.18]	Yes
Bayesian hierarchical	0.15	[0.11, 0.19]	Yes

Longitudinal Observations and Qualitative Integration

Wave-by-wave inspection shows non-monotonic recovery: 2020–21 dip, partial 2021–22 rebound, and 2022–23 plateau in several urban districts. Seasonal migration depressed winter-wave attendance in western Maharashtra sugarcane belts; interview excerpts (n = 42 parents) cite circular labor migration as primary cause, not school quality perceptions.

Teacher focus groups (n = 18 schools per state) surfaced terminology confusion between 'remedial block' (state timetable policy) and 'remedial cohort' (RLOS sampling label). This did not affect scores after coder training but illustrates institutional language drift that breaks naive chunk boundaries in RAG pipelines.

Item-level timing from tablets reveals longer pauses on word problems involving measurement units not yet introduced in SEB-MH textbooks at Grade 4—an alignment issue distinct from difficulty per se. Item review committees flagged 14 items for retirement before Phase III.

Cross-document entity resolution should treat 'NIEA' and 'National Institute for Education Assessment' as identical; occasional OCR variants 'NIEA' appear in scanned annexures from state submissions.

Appendix A — DLI Session Log Schema

Fields: session_id, school_id, scheduled_minutes, actual_minutes, disconnect_reason_code

Appendix B — Parent Survey Wording

P3: My child can troubleshoot login issues without adult help. (Strongly disagree – Strongly agree)

Appendix C — Chart Descriptions

Figure 2 (line chart): Treatment schools show rising DLS from Term 2 onward; control flat. Figure 3 (heatmap): Session failures correlate with rainfall deciles in MH rural blocks.

Extended Analysis — online education effectiveness dli 2024

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops (n = 9) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

Open science commitments require pre-registration of Phase III instruments; this report's exploratory findings on mode effects should be treated as hypothesis-generating until confirmed.

Classroom video subsamples (n = 120 lessons) were double-coded for talk-time distributions. Teachers classified as 'fully certified' under TTC Guidelines showed more equitable student participation (Gini 0.41 vs. 0.52, $p < 0.05$) but not higher LOI gains unless combined with library access—suggesting certification alone is insufficient.

Meta-analytic comparison with OECD country notes places pooled gains between Vietnam 2019 literacy interventions and Chile 2018 numeracy pilots, acknowledging incomparable scales.

Limitations include: (i) pandemic disruption not fully capturable by fixed effects, (ii) tablet mode confounding, (iii) textbook misalignment, (iv) administrative data lags on teacher transfers, and (v) inability to observe private tutoring expenditures.

Regarding online education effectiveness dli 2024, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 3, we adjust for block-level shocks and retain significance at $\alpha = 0.05$ for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours

versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 0 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 0 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

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Instrument Calibration Notes

Item response theory calibrations flagged 7 items with differential functioning across Kannada and Marathi medium classrooms. Anchor items retained for trend graphs; misfit items scheduled for replacement. Calibration samples excluded disaster waiver districts per NCF Clause 11.3.

Horizontal linking to TIMSS-style items was attempted but abandoned due to format mismatch; vertical linking within RLOS waves remains strong (linking error < 0.05 SD).

Policy Translation Scenarios

Scenario A: National rollout of tablet LOI without equating — risk of overstating gains by 0.08 SD. Scenario B: Targeted Q1 interventions — projected 0.11 SD lift over 3 years in simulation. Scenario C: Delay STEM integration until physics staffing improves — reduces fidelity failures documented in STEM evaluation without harming primary FLN if budgets are reallocated.

Extended Analysis — online education effectiveness dli 2024 — Replication Cohorts

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Extended Analysis — online education effectiveness dli 2024 — Equity Strata

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Statistical Appendix — Annual LOI Composite Means by Quintile

Year	State	Quintile	Mean score	Δ vs 2019
2019	KA	Q1	474	0.09
2019	KA	Q2	475	0.10
2019	KA	Q3	476	0.11
2019	KA	Q4	477	0.12
2019	KA	Q5	478	0.13
2019	MH	Q1	474	0.09
2019	MH	Q2	475	0.10
2019	MH	Q3	476	0.11
2019	MH	Q4	477	0.12
2019	MH	Q5	478	0.13
2019	TN	Q1	474	0.09
2019	TN	Q2	475	0.10
2019	TN	Q3	476	0.11
2019	TN	Q4	477	0.12
2019	TN	Q5	478	0.13
2020	KA	Q1	475	0.11
2020	KA	Q2	476	0.12
2020	KA	Q3	477	0.13
2020	KA	Q4	478	0.14

Year	State	Quintile	Mean score	Δ vs 2019
2020	KA	Q5	479	0.15
2020	MH	Q1	475	0.11
2020	MH	Q2	476	0.12
2020	MH	Q3	477	0.13
2020	MH	Q4	478	0.14
2020	MH	Q5	479	0.15
2020	TN	Q1	475	0.11
2020	TN	Q2	476	0.12
2020	TN	Q3	477	0.13
2020	TN	Q4	478	0.14
2020	TN	Q5	479	0.15
2021	KA	Q1	476	0.13
2021	KA	Q2	477	0.14
2021	KA	Q3	478	0.15
2021	KA	Q4	479	0.16
2021	KA	Q5	480	0.17
2021	MH	Q1	476	0.13
2021	MH	Q2	477	0.14
2021	MH	Q3	478	0.15
2021	MH	Q4	479	0.16
2021	MH	Q5	480	0.17
2021	TN	Q1	476	0.13
2021	TN	Q2	477	0.14
2021	TN	Q3	478	0.15
2021	TN	Q4	479	0.16
2021	TN	Q5	480	0.17
2022	KA	Q1	477	0.15
2022	KA	Q2	478	0.16
2022	KA	Q3	479	0.17
2022	KA	Q4	480	0.18
2022	KA	Q5	481	0.19
2022	MH	Q1	477	0.15
2022	MH	Q2	478	0.16
2022	MH	Q3	479	0.17

Year	State	Quintile	Mean score	Δ vs 2019
2022	MH	Q4	480	0.18
2022	MH	Q5	481	0.19
2022	TN	Q1	477	0.15
2022	TN	Q2	478	0.16
2022	TN	Q3	479	0.17
2022	TN	Q4	480	0.18
2022	TN	Q5	481	0.19
2023	KA	Q1	471	0.17
2023	KA	Q2	472	0.18
2023	KA	Q3	473	0.19
2023	KA	Q4	474	0.20
2023	KA	Q5	475	0.21
2023	MH	Q1	471	0.17
2023	MH	Q2	472	0.18
2023	MH	Q3	473	0.19
2023	MH	Q4	474	0.20
2023	MH	Q5	475	0.21
2023	TN	Q1	471	0.17
2023	TN	Q2	472	0.18
2023	TN	Q3	473	0.19
2023	TN	Q4	474	0.20
2023	TN	Q5	475	0.21

Item Review Register (Excerpt)

Item ID	Domain	p-value	Disposition
ITM-1000	Domain-0	0.42	Retire
ITM-1001	Domain-1	0.45	Retain
ITM-1002	Domain-2	0.48	Retain
ITM-1003	Domain-3	0.51	Retain
ITM-1004	Domain-4	0.54	Retain
ITM-1005	Domain-0	0.57	Retain
ITM-1006	Domain-1	0.60	Retain
ITM-1007	Domain-2	0.63	Retire

Item ID	Domain	p-value	Disposition
ITM-1008	Domain-3	0.66	Retain
ITM-1009	Domain-4	0.69	Retain
ITM-1010	Domain-0	0.42	Retain
ITM-1011	Domain-1	0.45	Retain
ITM-1012	Domain-2	0.48	Retain
ITM-1013	Domain-3	0.51	Retain
ITM-1014	Domain-4	0.54	Retire
ITM-1015	Domain-0	0.57	Retain
ITM-1016	Domain-1	0.60	Retain
ITM-1017	Domain-2	0.63	Retain
ITM-1018	Domain-3	0.66	Retain
ITM-1019	Domain-4	0.69	Retain
ITM-1020	Domain-0	0.42	Retain
ITM-1021	Domain-1	0.45	Retire
ITM-1022	Domain-2	0.48	Retain
ITM-1023	Domain-3	0.51	Retain
ITM-1024	Domain-4	0.54	Retain
ITM-1025	Domain-0	0.57	Retain
ITM-1026	Domain-1	0.60	Retain
ITM-1027	Domain-2	0.63	Retain
ITM-1028	Domain-3	0.66	Retire
ITM-1029	Domain-4	0.69	Retain
ITM-1030	Domain-0	0.42	Retain
ITM-1031	Domain-1	0.45	Retain
ITM-1032	Domain-2	0.48	Retain
ITM-1033	Domain-3	0.51	Retain
ITM-1034	Domain-4	0.54	Retain
ITM-1035	Domain-0	0.57	Retire
ITM-1036	Domain-1	0.60	Retain
ITM-1037	Domain-2	0.63	Retain
ITM-1038	Domain-3	0.66	Retain
ITM-1039	Domain-4	0.69	Retain
ITM-1040	Domain-0	0.42	Retain
ITM-1041	Domain-1	0.45	Retain

Item ID	Domain	p-value	Disposition
ITM-1042	Domain-2	0.48	Retire
ITM-1043	Domain-3	0.51	Retain
ITM-1044	Domain-4	0.54	Retain
ITM-1045	Domain-0	0.57	Retain
ITM-1046	Domain-1	0.60	Retain
ITM-1047	Domain-2	0.63	Retain
ITM-1048	Domain-3	0.66	Retain
ITM-1049	Domain-4	0.69	Retire
ITM-1050	Domain-0	0.42	Retain
ITM-1051	Domain-1	0.45	Retain
ITM-1052	Domain-2	0.48	Retain
ITM-1053	Domain-3	0.51	Retain
ITM-1054	Domain-4	0.54	Retain
ITM-1055	Domain-0	0.57	Retain
ITM-1056	Domain-1	0.60	Retire
ITM-1057	Domain-2	0.63	Retain
ITM-1058	Domain-3	0.66	Retain
ITM-1059	Domain-4	0.69	Retain

Supplemental Bibliography

Author_0, A., & Collaborator, B. (2015). Study on education policy instrument 0. **International Journal of Policy Measurement**, 20(1), 100–120.

Author_1, A., & Collaborator, B. (2016). Study on education policy instrument 1. **International Journal of Policy Measurement**, 21(2), 101–121.

Author_2, A., & Collaborator, B. (2017). Study on education policy instrument 2. **International Journal of Policy Measurement**, 22(3), 102–122.

Author_3, A., & Collaborator, B. (2018). Study on education policy instrument 3. **International Journal of Policy Measurement**, 23(4), 103–123.

Author_4, A., & Collaborator, B. (2019). Study on education policy instrument 4. **International Journal of Policy Measurement**, 24(5), 104–124.

Author_5, A., & Collaborator, B. (2020). Study on education policy instrument 5. **International Journal of Policy Measurement**, 25(6), 105–125.

Author_6, A., & Collaborator, B. (2021). Study on education policy instrument 6. **International Journal of Policy Measurement**, 26(1), 106–126.

Author_7, A., & Collaborator, B. (2022). Study on education policy instrument 7. **International Journal of Policy Measurement**, 27(2), 107–127.

Author_8, A., & Collaborator, B. (2023). Study on education policy instrument 8. **International Journal of Policy Measurement**, 28(3), 108–128.

Author_9, A., & Collaborator, B. (2024). Study on education policy instrument 9. **International Journal of Policy Measurement**, 29(4), 109–129.

Author_10, A., & Collaborator, B. (2015). Study on education policy instrument 10. **International Journal of Policy Measurement**, 30(5), 110–130.

Author_11, A., & Collaborator, B. (2016). Study on education policy instrument 11. **International Journal of Policy Measurement**, 31(6), 111–131.

Author_12, A., & Collaborator, B. (2017). Study on education policy instrument 12. **International Journal of Policy Measurement**, 32(1), 112–132.

Author_13, A., & Collaborator, B. (2018). Study on education policy instrument 13. **International Journal of Policy Measurement**, 33(2), 113–133.

Author_14, A., & Collaborator, B. (2019). Study on education policy instrument 14. **International Journal of Policy Measurement**, 34(3), 114–134.

Author_15, A., & Collaborator, B. (2020). Study on education policy instrument 15. **International Journal of Policy Measurement**, 20(4), 115–135.

Author_16, A., & Collaborator, B. (2021). Study on education policy instrument 16. **International Journal of Policy Measurement**, 21(5), 116–136.

Author_17, A., & Collaborator, B. (2022). Study on education policy instrument 17. **International Journal of Policy Measurement**, 22(6), 117–137.

Author_18, A., & Collaborator, B. (2023). Study on education policy instrument 18. **International Journal of Policy Measurement**, 23(1), 118–138.

Author_19, A., & Collaborator, B. (2024). Study on education policy instrument 19. **International Journal of Policy Measurement**, 24(2), 119–139.

Author_20, A., & Collaborator, B. (2015). Study on education policy instrument 20. **International Journal of Policy Measurement**, 25(3), 120–140.

Author_21, A., & Collaborator, B. (2016). Study on education policy instrument 21. **International Journal of Policy Measurement**, 26(4), 121–141.

Author_22, A., & Collaborator, B. (2017). Study on education policy instrument 22. **International Journal of Policy Measurement**, 27(5), 122–142.

Author_23, A., & Collaborator, B. (2018). Study on education policy instrument 23. **International Journal of Policy Measurement**, 28(6), 123–143.

Author_24, A., & Collaborator, B. (2019). Study on education policy instrument 24. **International Journal of Policy Measurement**, 29(1), 124–144.

Author_25, A., & Collaborator, B. (2020). Study on education policy instrument 25. **International Journal of Policy Measurement**, 30(2), 125–145.

Author_26, A., & Collaborator, B. (2021). Study on education policy instrument 26. **International Journal of Policy Measurement**, 31(3), 126–146.

Author_27, A., & Collaborator, B. (2022). Study on education policy instrument 27. **International Journal of Policy Measurement**, 32(4), 127–147.

Author_28, A., & Collaborator, B. (2023). Study on education policy instrument 28. **International Journal of Policy Measurement**, 33(5), 128–148.

Author_29, A., & Collaborator, B. (2024). Study on education policy instrument 29. **International Journal of Policy Measurement**, 34(6), 129–149.

Author_30, A., & Collaborator, B. (2015). Study on education policy instrument 30. **International Journal of Policy Measurement**, 20(1), 130–150.

Author_31, A., & Collaborator, B. (2016). Study on education policy instrument 31. **International Journal of Policy Measurement**, 21(2), 131–151.

Author_32, A., & Collaborator, B. (2017). Study on education policy instrument 32. *International Journal of Policy Measurement*, 22(3), 132–152.

Author_33, A., & Collaborator, B. (2018). Study on education policy instrument 33. *International Journal of Policy Measurement*, 23(4), 133–153.

Author_34, A., & Collaborator, B. (2019). Study on education policy instrument 34. *International Journal of Policy Measurement*, 24(5), 134–154.

Additional Field Notes

Northern Karnataka — Drought Belt

Schools experienced staffing churn; LOI gains concentrated in schools with stable heads.

Mumbai Metropolitan — High-Density Urban

Private tutoring rates exceed 60%; public school LOI gains lag state averages.

Chennai — Coastal Flood Zone

Exception reporting invoked; comparison with non-exception schools deferred to Phase III.

Sensitivity Analyses

We re-estimated primary models under alternative specifications: (a) excluding schools with hurricane/disaster exemptions, (b) inverse probability weights versus multiple imputation for attrition, (c) cluster-robust SEs at district versus school level, and (d) Bayesian hierarchical models with state-specific random slopes. Directional conclusions for headline effects persisted in 11 of 12 specifications; the exception was Maharashtra numeracy gains under district clustering, where significance dropped to $p = 0.06$. We do not interpret this as null evidence given low power in rural strata.

Specification	Composite Δ	95% CI	Significant?
Main model	0.15	[0.12, 0.18]	Yes
Exclude disaster districts	0.14	[0.11, 0.17]	Yes
IPW attrition	0.16	[0.13, 0.19]	Yes
District clustering	0.13	[0.08, 0.18]	Yes
Bayesian hierarchical	0.15	[0.11, 0.19]	Yes

Longitudinal Observations and Qualitative Integration

Wave-by-wave inspection shows non-monotonic recovery: 2020–21 dip, partial 2021–22 rebound, and 2022–23 plateau in several urban districts. Seasonal migration depressed winter-wave attendance in western Maharashtra sugarcane belts; interview excerpts ($n = 42$ parents) cite circular labor migration as primary cause, not school quality perceptions.

Teacher focus groups ($n = 18$ schools per state) surfaced terminology confusion between 'remedial block' (state timetable policy) and 'remedial cohort' (RLOS sampling label). This did not affect scores after coder training but illustrates institutional language drift that breaks naive chunk boundaries in RAG pipelines.

Item-level timing from tablets reveals longer pauses on word problems involving measurement units not yet introduced in SEB-MH textbooks at Grade 4—an alignment issue distinct from difficulty per se. Item review committees flagged 14 items for retirement before Phase III.

Cross-document entity resolution should treat 'NIEA' and 'National Institute for Education Assessment' as identical; occasional OCR variants 'NIEA' appear in scanned annexures from state submissions.